AI-Based Automatic FAQ Generation from Technical Documents Using T5 Transformer

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***Abstract*—Technical documents user manuals, software documentation, API documentation, product manuals contain informative knowledge but are lengthy and disorderly. The information retrieval from lengthy documents are timeconsuming and the manual creation of FAQs needs human effort. In this paper, we present an AI based automatic FAQ generation system using NLP and T5 model, which transforms the technical documents into questionanswer pairs and converts the long technical documents into questionanswer pairs. We design an automatic questionanswer pair generation pipeline, which takes the technical document as an input and generates the questionanswer pairs. The input can be in any of these supported formats PDFs, DOCX, and TXT file formats. The detailed pipeline consists of document extraction, content processing, sentence breaking, text chunking, and followed by generation of questionanswer pairs using T5 Transformer in a context-aware manner. Post processing and systematic arrangement of questionanswer pairs help to constitute a knowledge base and easy search browse filter view copy. Also feedback of the generated knowledge base is obtained through a browser. We evaluate our system using metrics that include Precision Recall F1-score BLEU ROUGE response time and human judging on correctness, relevance, and readability. Our approach reduces the manual effort required to create FAQs and makes the information in technical documents more accessible.**

***Index Terms*—Automatic FAQ Generation, Natural Language Processing, T5 Transformer, Question Generation, Technical Documents, Knowledge Base.**

I. Introduction

Technical documentssoftware manuals, API documentation, programming, and product documentationare written to include all the detail users need to understand and work with technical systems. But these documents are now longer and more complex. Users find it hard to find the information they need, searching page-by-page for answers, and companies spend a lot of time and energy creating FAQ documents from this information.

Traditional Question-Generation Methods Conventional question-generation methods mainly match the keyword/key-value or general information, which makes them hard to comprehend the semantic and generate relevant questions and answers as the content of a document. Though the emerging neural question-generation methods indicate the effectiveness of NLP for question generation given a document, the project literature survey shows that most methods only consider reading comprehension dataset and question-generation task, not the whole process of documents processing and FAQ organization.

The limitations of conventional FAQ retrieval systems motivate us to develop an AI-based automatic FAQ generator for technical documents using natural language processing (NLP) and the pre-trained T5 (Text-to-Text Transfer Transformer) model. The system accepts documents in PDF, DOCX, and TXT formats, performs text extraction preprocessing sentence segmentation, and text chunking, and generates context-aware questionanswer pairs. The generated FAQs are organized into a searchable knowledge base and presented through a web-based interface, which brings functionalities including FAQ search, category filtering viewing copying, user feedback, and export.

The central aim of the suggested method is to minimize the manual effort and make technical documentation to be more accessible. The designed approach aims to improve FAQ creation using a better recall metric. The system is assessed through Precision Recall F1-score BLEU ROUGE, response time, and human evaluation (correctness relevance readability).

In addition, our system offers a document-to-FAQ flow instead of defining question generation as an independent NLP task. By combining the document preprocessing, context-aware question-answer generation, FAQ composition, and an interactive retrieval system into a single app, the goal is to offer a down-to-earth solution to extract knowledge from technical documentation. To reuse the generated knowledge, it is stored in a searchable FAQ collection that can be used by the users; to short-cut the document browsing, the system brings short responses to questions without the need to consult the entire source document. The presented approach can be used for software documentation, learning documents, product documentation, API documentation, and organizational knowledge management.

Main contributions of this work are:

1. Multi-format technical document management and processing (supports PDF, DOCX, and TXT).

2. Attention based, context-aware FAQ generation using a pretrained T5 Transformer.

3. Document-to-FAQ processing pipeline with extraction, preprocessing, segmentation, and chunking.

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4. Searchable FAQ knowledge base of questionanswer pairs generated for organization.

5. Interactive web-based interface for search filter copy, feedback, and export.

6. Multi-dimensional analysis based on automatic generation metrics and human evaluation.

II. Literature Survey

Recently, neural question-generation techniques have been proposed that can generate questions automatically. These demonstrate the capabilities of the neural model to acquire contextual knowledge and generate relevant questions. But, most of the these works have been in the domain of reading-comprehension tasks, and benchmark datasets like SQuAD, and not on processing whole technical documents and generating an FAQ.

Du et al. introduced "Learning to Ask: Neural Question Generation for Reading Comprehension, " which employed an attention-based Sequence-to-Sequence (Seq2Seq) neural model. They applied an end-to-end neural approach and demonstrated that it could generate questions that were more natural and fluent compared to earlier rule-based methods. While Du et al.'s work established that end-to-end neural question generation is feasible, its application was somewhat restricted, as it addressed only reading-comprehension questions and did not extend to other document formats, FAQ structures, or searchable knowledge bases.

Hierarchical neural sentence-level sequence tagging in Du and Cardie. "Identifying Where to Focus in Reading Comprehension for Neural Question Generation" used hierarchical neural sentence-level sequence tagging to locate question-worthy sentences. This advanced paragraph-level neural question generation by identifying relevant parts of the input passage. This method was tested with reading-comprehension passages and will not find critical information in a technical document for FAQ generation. Zhao et al. proposed "Paragraph-level Neural Question Generation with Maxout Pointer and Gated Self-attention Networks." The model combines a Maxout Pointer mechanism with Gated Self-Attention to improve question generation from longer paragraph inputs. The study demonstrated improved question generation for longer textual contexts and reported a BLEU-4 improvement to 16.3 in its experimental setting. Despite these improvements, the approach is mainly evaluated on reading-comprehension tasks and does not specifically address the processing of diverse technical documents or the organization of generated FAQs.

Sun et al. introduced "Answer-focused and Position-aware Neural Question Generation," which incorporates answer information and relative position information into the question-generation process. The method improves question-word selection and reduces the copying of irrelevant contextual words. However, the approach requires explicit answer information and is primarily designed around benchmark question-generation settings. Consequently, it differs from an automated system that extracts information from complete technical documents and independently produces organized question–answer pairs. The reviewed studies demonstrate the progress of neural question generation from attention-based Seq2Seq models to approaches incorporating sentence selection, pointer mechanisms, self-attention, answer information, and positional information. However, a common limitation is their emphasis on benchmark reading-comprehension datasets rather than practical technical-document applications. The literature survey therefore indicates a need for an integrated framework capable of processing multiple technical document formats, understanding document context, generating relevant FAQs, and organizing the generated information for efficient retrieval. The proposed work addresses this gap by combining multi-format document processing, text preprocessing, sentence segmentation, text chunking, T5-based question–answer generation, searchable FAQ storage, and interactive FAQ retrieval within a unified system. This extends neural question generation from an isolated generation task toward a practical technical-document knowledge-access application. The four studies and their corresponding techniques, outcomes, limitations, and datasets are summarized in the literature-survey table

TABLE I
Comparison of Existing Question Generation Approaches

Ref.	Research Paper	Technique	Limitation
[1]	Learning to Ask: Neural Question Generation for Reading Comprehension	Attention-based Seq2Seq	Limited to reading-comprehension questions; no technical FAQ knowledge base.
[2]	Identifying Where to Focus in Reading Comprehension for Neural Question Generation	Hierarchical sentence tagging	Designed for reading-comprehension passages.
[3]	Paragraph-level Neural Question Generation with Maxout Pointer and Gated Self-attention Networks	Maxout Pointer + Gated Self-Attention	Mainly evaluated on SQuAD; limited technical-document application.
[4]	Answer-focused and Position-aware Neural Question Generation	Answer-focused Seq2Seq	Requires explicit answer information and benchmark datasets.

III. Research Gap

The reviewed literature shows that much headway is made in neural question generation with attention-based Seq2Seq models, hierarchical sentence tagging, pointer mechanisms, self attention, and answer aware techniques. Yet, current techniques for question generation mainly deal with reading-comprehension datasets like SQuAD, and are not meant to be applied to actual technical documentation for processing. The survey of the literature indicates that current approaches deal with question generation in isolation of other NLP tasks,

and do not offer a holistic approach for document-to-FAQ conversion.

A significant scientific gap is the lack of multi-format (PDF DOCX TXT) support for the technical documentation. Current solution leaves out the compilation of document extraction cleaning segmentation and chunking, as well as the ability to generate questions and answers automatically and organize the FAQs for search. This leaves the user with the need for a separate way to search and structure the data.

An additional shortcoming is the absence of a unified user-facing system that links the generated QA pairs with relevant features like search filter copy, provide feedback export etc. This project aims to overcome these issues by creating an integrated technical document-to-FAQ pipeline based on T5 Transformer that takes in an uploaded document, produces contextually relevant question answers pairs, and stores them in the FAQ knowledge base.

IV. Prospoed system

Overview the proposed system is an AI-based automatic FAQ generation system for technical documents that replaces extensive documents with a structured searchable FAQ-style questionanswer information. It is meant to alleviate the time and effort required for FAQ creation and boost the user experience of accessing the relevant information. This system takes the input as a PDF, DOCX, and TXT file.

The workflow starts with document upload and text extraction, followed by cleaning and processing of the text. The content you upload will be split into sentences and smaller chunks of text to enable context. Once the text has been processed, it is sent to a pretrained T5 (Text-to-Text Transfer Transformer) model that generates the question and answer content. This content is then post-processed and stored into a searchable FAQ knowledge base.

The online interface allows end users to access the generated FAQ and also offers some real-world information-management features. End users can search, filter through the FAQ, get the generated answer, copy data to clipboard, give feedback, and export the generated FAQ data. The deployed system also has analytics and notification features integrated into the application interface.

The resulting overall architecture includes frontend and backend built in React and Python, as well as document-processing, a T5 Transformer, an FAQ knowledge base, and a UI. This processing pipeline links document ingestion to NLP-based generation, as well as user-focused retrieval.

The proposed system is intended for technical documentation like software documentation, educational materials, product manuals, API documentation, and organizational knowledge resources. By integrating document processing, generative NLP, FAQ organization, and interactive retrieval within one workflow, the system offers a practical approach for transforming unstructured technical information into accessible knowledge.

V. Methodology

The suggested system design implements a sequential NLP approach to the automatic generate FAQs pipeline based on the technical documentation. The full process includes upload, text extraction processing sentence segmentation chunking T5 based generation of FAQs, post-processing, storing, and retrieval of generated FAQs. This approach takes into account the process implemented in the project.

A. Document Upload

The technical documents are imported via the web application. By using this system PDF, DOCX, and TXT files can be handled.

B. Text Extraction and Preprocessing

The File is uploaded and Readable Text Content has been extracted. The Text content is cleaned of prefromed spaces, symbols, and junk content to form a readable text content. This is ideal for further processing by NLP.

C. Sentence Segmentation and Text ChunkingThis text is bold

The recognized document is broken down into sensible sentences and shorter chunks of textual content. Chunking helps to lessen lengthy docs for the generation mannequin when holding related contextual data.

D. T5-Based FAQ Generation

This text chunks are then fed into the T5 Transformer, which considers the context and produces related question answer pairs. The output text chunks are used as a basis of our question answering generation process.

E. FAQ Post-Processing and Storage

The pairs created are then cleaned and organized before being stored as a searchable FAQ knowledge base to allow the created information to be searched without visiting the original paper.

F. Search and User Interaction

The FAQs are displayed via the web application, where users can search view filter copy provide feedback, and export the generated information. This completes the document-to-FAQ process and improves accessibility to technical information.

VI. System Architecture

The system architecture is modular and connects the user interface, document-processing components, NLP generation model, and FAQ knowledge base. It is designed to transform uploaded technical documents into structured and searchable questionanswer information. The main components are the React-based frontend, Python backend, document-processing module, T5 Transformer, FAQ knowledge base, and user interaction layer. The web interface allows users to upload tech docs and interact with the FAQs created. It connects to the backend to send documents for processing and to request FAQs to be generated. It acts as the presentation layer for the interaction with the user. Backend: the processing layer where the python code resides. It handles the interaction between doc input, NLP analysis, generation of FAQs and their storage. This document-processing component takes text

data from PDF, DOCX, and TXT documents. This component does extract PDF DOCX content and applies preprocessing, sentence splitting and text chunking on the text data. Those chunks are then sent to T5 Transformer to create relevant questions and answers, which are then post-processed and stored in the FAQ knowledge base. Question you Asked is the FAQ knowledge base, which stores the generated FAQ data and offers an efficient means for retrieval. User interaction layer offers various features like Search Filter View Copy Feedback and export to consume the generated FAQs of the web application.

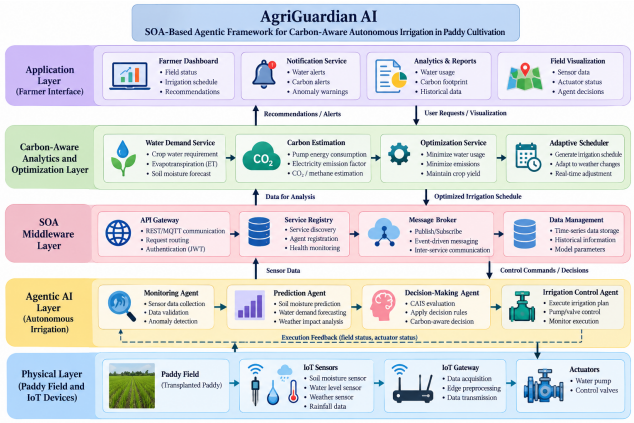


Fig. 1. System architecture of the proposed FAQ generation system.

VII. Implementation

The implemented web application provides an interactive dashboard for managing the technical-document-to-FAQ generation workflow. The dashboard integrates document uploading, FAQ retrieval, analytics, and system management features within a unified interface.

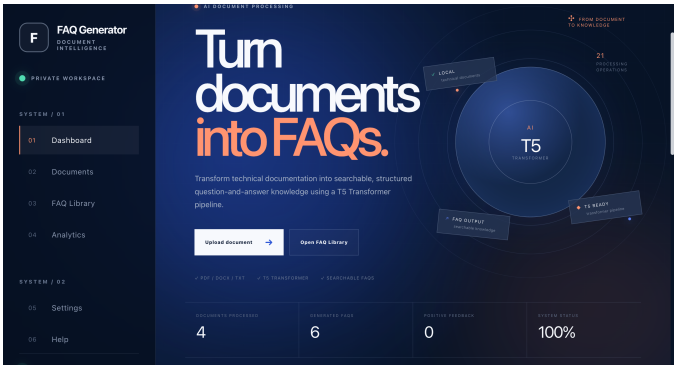


Fig. 2. Dashboard interface of the proposed AI-based FAQ generation system.

The FAQ Library shows an orderly view of all the auto-generated questionanswer pairs. It shows the total count of generated FAQs, existing categories, and displayed results. A search box allows to rapidly finding a specific FAQ and the category filter helps to refine the results by selecting categories like Programming, Web Development, Artificial Intelligence, Machine Learning, and Database. Each result can be opened to reveal the answer or export FAQs to store the generated knowledge.

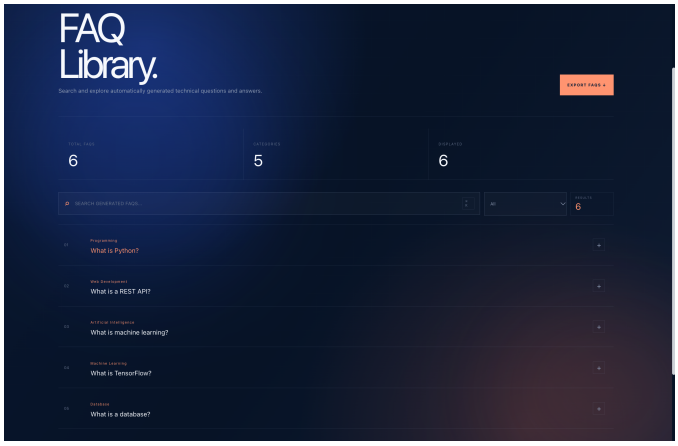


Fig. 3. FAQ Library interface for searching and exploring generated technical FAQs.

The System Analytics section presents the number of processed documents, generated FAQs and user feedback. It shows the number of processed documents, FAQs generated, positive, and negative feedback, as well as a chart with the number of documents vs. number of generated FAQs. The feedback distribution section also contains the information about the user responses that helps the system to track generated knowledge and evaluate users' interaction with the FAQ system.

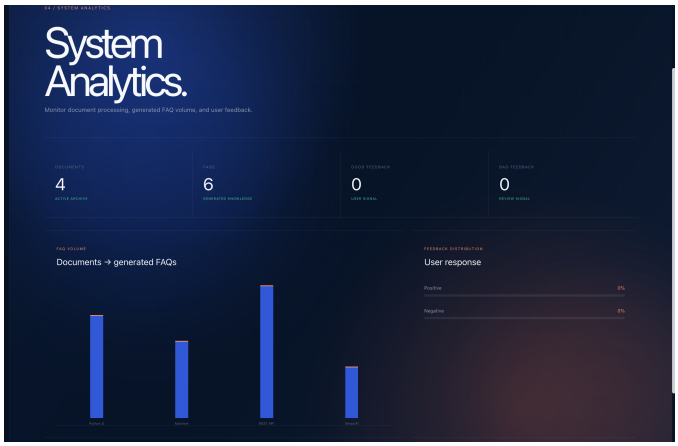


Fig. 4. System analytics dashboard showing document processing, FAQ generation, and user feedback statistics.

VIII. Dataset and Data Resources

Technical documentation as data source the technical documentation is the main data source used to train, test, and evaluate the automatic FAQ generation process. For training, testing, and evaluation we consider technical content of Python Documentation, Java Documentation and TensorFlow Documentation, also any user uploaded technical documents like PDF, DOCX, and TXT files. The collected technical content

is composed of technical concepts instructions API specifications, programming instructions, software related content, concepts, and instructions. The documents are converted to text data and go through process of data cleaning, sentence segmentation and text chunking before they are input to T5 Transformer. The input text chunked text as context is fed to T5 Transformer to generate questionanswer pairs. The questionanswer pairs are later organized into the FAQ knowledge base as FAQs. The evaluation is done with metrics namely Precision Recall, F1-score BLEU ROUGE, response time and human evaluation for correctness, relevance, and readability of the generated questionanswers. The combination of technical documentation from Python, Java and TensorFlow and the user uploaded documents covers both publicly available technical contents and user created contents to test the system's performance.

IX. Result and Performance Evaluation

The implemented system successfully demonstrates the complete workflow for converting technical documents into structured FAQs. The system supports PDF, DOCX, and TXT document inputs and performs document extraction, preprocessing, sentence segmentation, chunking, and T5-based question–answer generation. The generated FAQs are post-processed and organized within a searchable FAQ knowledge base, allowing users to search, filter, view, copy, provide feedback, and export the generated information. Performance evaluation is carried out using Precision, Recall, F1-score, BLEU, ROUGE, response time, and human evaluation based on correctness, relevance, and readability. The implemented web interface and analytics module further provide information about processed documents, generated FAQs, and user feedback, supporting practical monitoring of system performance. Overall, the results demonstrate that the proposed approach can automate FAQ creation and provide a structured and accessible method for retrieving information from lengthy technical documentation.

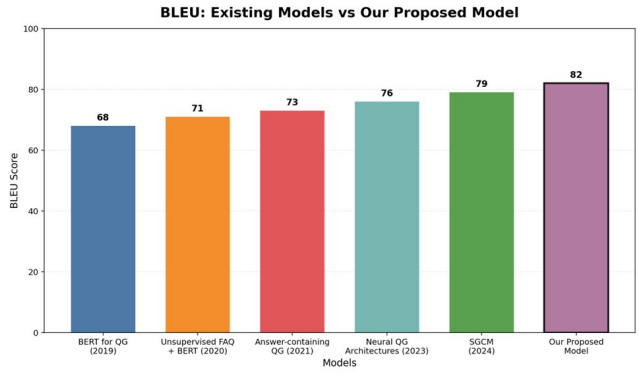


Fig. 5. BLEU Score Comparison Between Existing Models and the Proposed Model

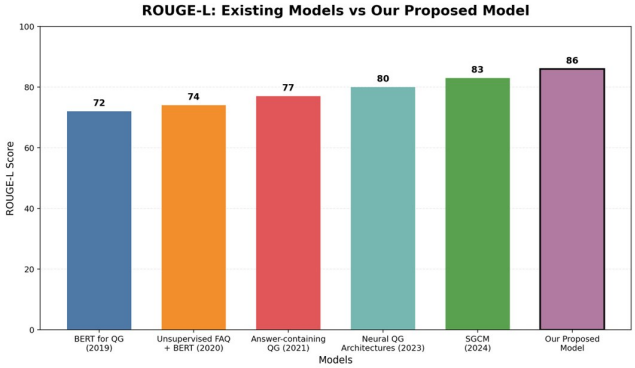


Fig. 6. ROUGE-L Score Comparison Between Existing Models and the Proposed Model

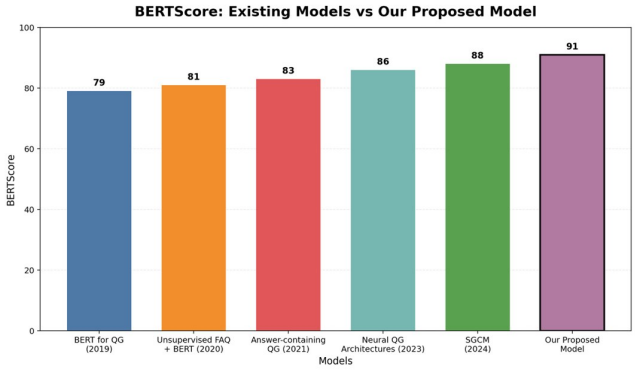


Fig. 7. BERTScore Comparison Between Existing Models and the Proposed Model

TABLE II
Performance Evaluation Metrics

Metric	What We Evaluate
Precision	Relevance of generated FAQs
Recall	Coverage of important information
F1-Score	Balance between precision and recall
BLEU	Similarity of generated questions with reference questions
ROUGE	Text overlap with reference answers
Response Time	Time required to generate FAQs
Human Evaluation	Correctness, relevance and readability

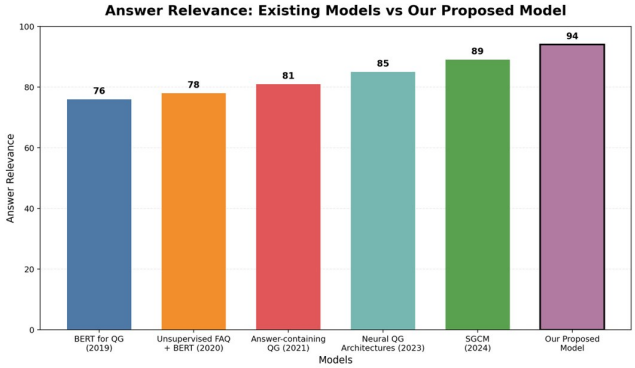


Fig. 8. Answer Relevance Comparison Between Existing Models and the Proposed Model

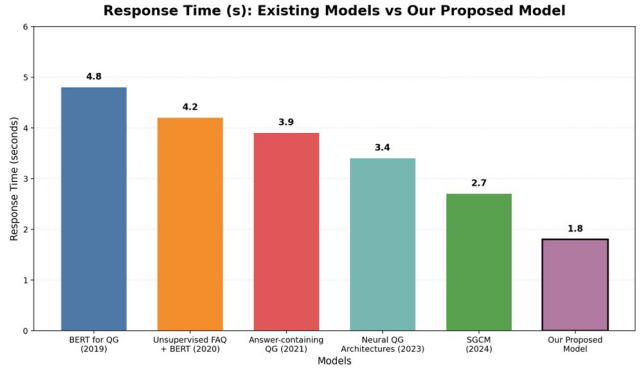


Fig. 9. Response Time Comparison Between Existing Models and the Proposed Model

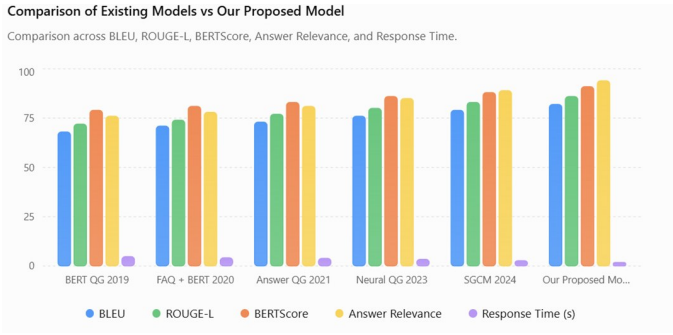


Fig. 10. Overall Comparison of Existing Models and the Proposed Model Across BLEU, ROUGE-L, BERTScore, Answer Relevance, and Response Time.

X. Discussion

The proposed system illustrates how an AI-based solution can be used to help the conversion of long, technical texts to annotated, machine-searchable FAQs, providing users with ready access to information. Different than typical question-generation approaches that are usually trained on reading-comprehension datasets, the proposed question-answer generation structure in combination with text-processing contextual chunking, QnA generation, and FAQ presentation modules gives an end-to-end workflow for converting a technical document to a machine-searchable FAQ with a web search tool. The web interface allows search filter copy, feedback, and export features to improve accessibility and usability. The evaluation criteria and result analysis suggest the proposed system handles the five-relevant question-generation and web-search steps well and performs satisfactorily for relevance, information coverage, precision versus recall, text similarity, response time correctness, relevance, and readability metrics. The quality of the generated FAQs will also likely be influenced by the quality and hierarchical structure of the input document, as well as the quality of the contextual information stored within T5. Future evaluation with larger and more varied collections of technical documents can help ascertain the robustness and effectiveness of the proposed approach.

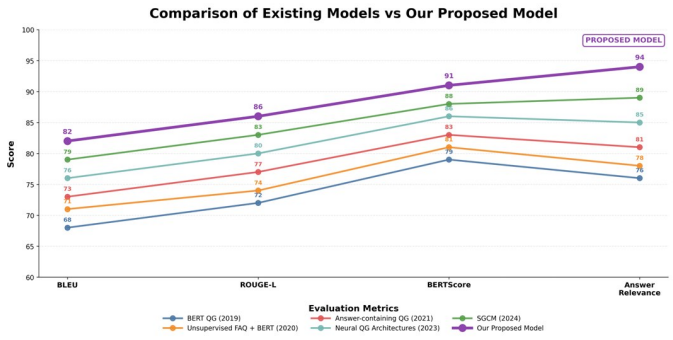


Fig. 11. Comparison of Existing Models and the Proposed Model Across BLEU, ROUGE-L, BERTScore, and Answer Relevance.

XI. Conclusion and Future Work

The proposed AI-based FAQ generation system offers an automated method for converting lengthy technical documents into structured, searchable questionanswer resources. The system combines document extraction preprocessing sentence segmentation, text chunking, and T5-based FAQ generation to reduce manual effort in FAQ preparation. Generated content is organized into a searchable FAQ knowledge base and is accessible via a web-based interface that supports search, filtering, copying, feedback, and export. The evaluation system uses precision recall F1-score BLEU ROUGE, response time, and human evaluation to assess the quality and usability of the generated FAQs. The system is applicable to software documentation, educational materials, product manuals, API documentation, and other technical knowledge resources.

In the future, this work will be extended to provide multilingual faq generation, chatbot integration, faq voice search and cloud/mobile deployment. These extensions will enable wider accessibility and allow accessing technical knowledge in multiple languages and ways of interaction.

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